Coincident Anomalies in

Ten Years of Photic Zone Profiles

An Interpretive Framework for OOI Shallow Profiler Data

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Three OOI Regional Cabled Array shallow profilers have produced continuous 9-per-day profiles of the upper 200 m since 2015 — a decade-long synoptic characterization of the Northeast Pacific photic zone. We present visualization, analysis, and an interpretive framework for these 15-parameter time series (CTD, bio-optical, biogeochemical), with a reproducible data-cleaning workflow. The framework centers on three forms of **anomaly coincidence**: **self** (a single sensor, repeated across profiles), **inter-sensor** (across multiple sensors), and **corroborated** (confirmed by external sources — deep profilers, circulation models, remote sensing, Argo). Anomalies are defined against per-sensor mean±std baseline envelopes over seasonal/annual/multi-year windows (published with the tools). We show self- and inter-sensor coincident signals at/near the pycnocline interpreted as internal waves and coherent water-mass emplacement persisting over multiple tidal cycles. Published as an open-science GitHub repository with a step-by-step acquisition→analysis workflow. E.g. buoyancy frequency $N^2 = -\frac{g}{\rho_0}\frac{\partial \rho}{\partial z}$.

Scan for the online poster & the argosy repository

1 · Baseline envelopes & the anomaly definition

[figure placeholder — per-sensor mean±std envelope over a time window]

One-line caption stub. Anomalies defined vs. per-sensor mean±std baselines over seasonal / annual / multi-year windows (published with the tools).

2 · Self & inter-sensor coincidence

[figure placeholder — self- and inter-sensor coincident anomalies]

One-line caption stub. Self-coincidence (one sensor, repeated across profiles) and inter-sensor coincidence (anomalies registering across multiple sensors).

3 · Pycnocline signals: internal waves & water-mass emplacement

[figure placeholder — pycnocline coincident signal over tidal cycles]

One-line caption stub. Coincident signals at/near the pycnocline interpreted as internal waves and coherent water-mass emplacement over multiple tidal cycles.

4 · Corroborated coincidence + open, reproducible science

[figure placeholder — external corroboration + cloud/open-science]

One-line caption stub. Corroboration from external sources (deep profilers, circulation models, remote sensing, Argo) and the cloud/open-science workflow (public S3, DOI-archived releases, step-by-step GitHub repo).